

Organometallic Chemistry

Ionic bonds, Covalent bond, Lewis Structures, Octate rule, Exceptions, VSEPR theory: predicting molecular geometries.

Valence bond theory and its limitations

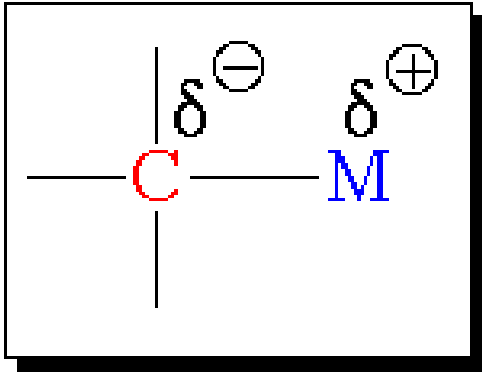
Crystal field theory, High- and low-spin states, Spectrochemical series, Tetragonal distortion of octahedral complexes

Applications of crystal field theory, Color and magnetic properties of transition metal complexes

Structure and reactivity of organometallic compounds,

Importance of organometallic chemistry in metal purification and in catalysis

What all compounds are considered as organometallic?



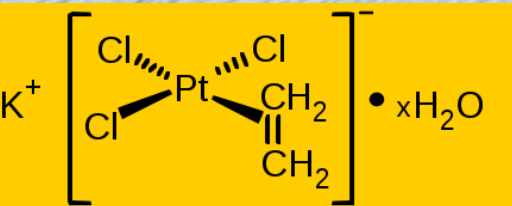
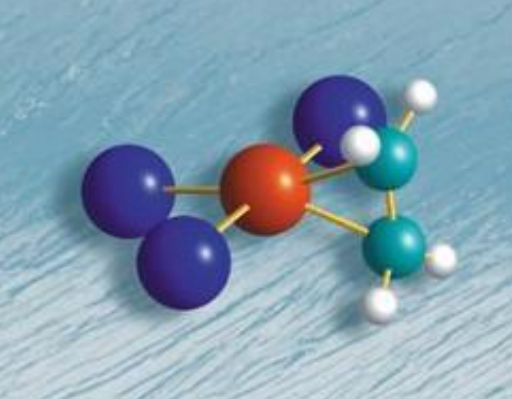
C always more electronegative compared to M

The leading journals of the field define an "organometallic" compound as one in which there is a **bonding interaction (ionic or covalent, localized or delocalized)** between one or more carbon atoms of an organic group or molecule and a main group, transition, lanthanide, or actinide metal atom (or atoms).

Following longstanding tradition, organic derivatives of the metalloids such as **boron, silicon, germanium, arsenic, and tellurium** also are included in this definition.

It is also understood that the element to which carbon is bound is **more electropositive than carbon** in organometallic chemistry.

*Traditional chemists do not agree for classifying **metal cyanide complexes** as organometallic*

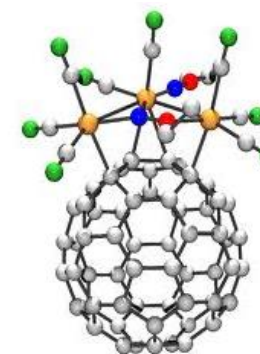
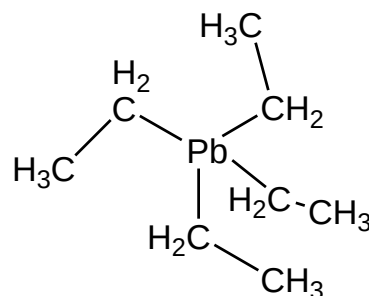
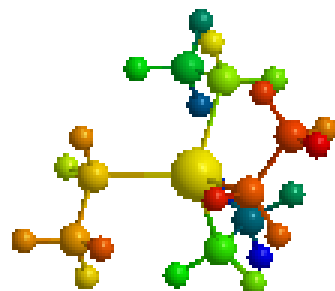
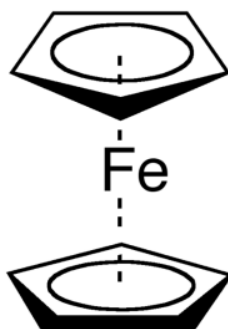
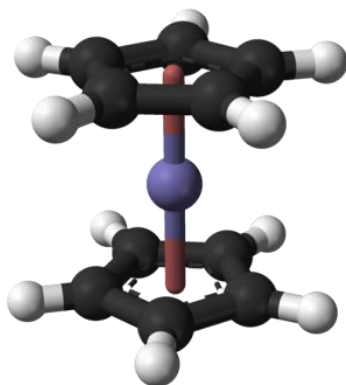
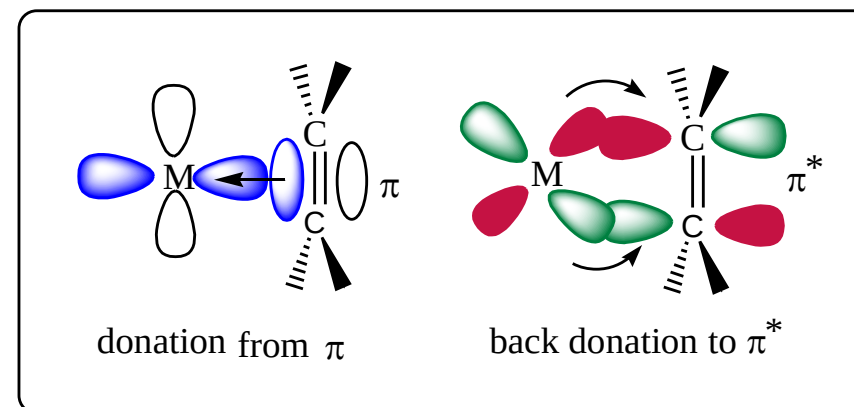
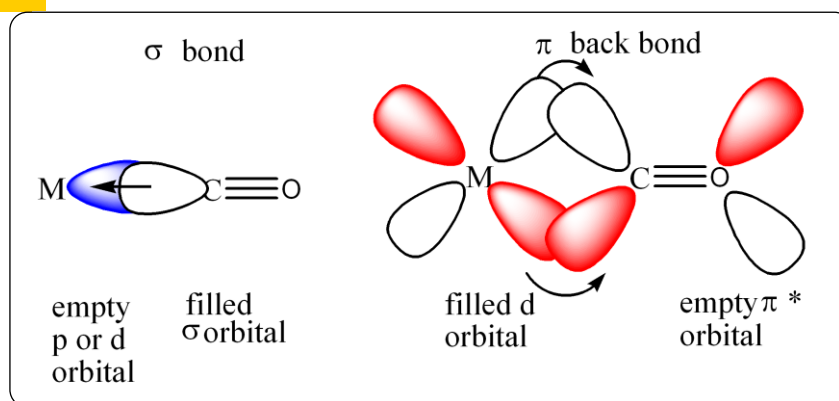
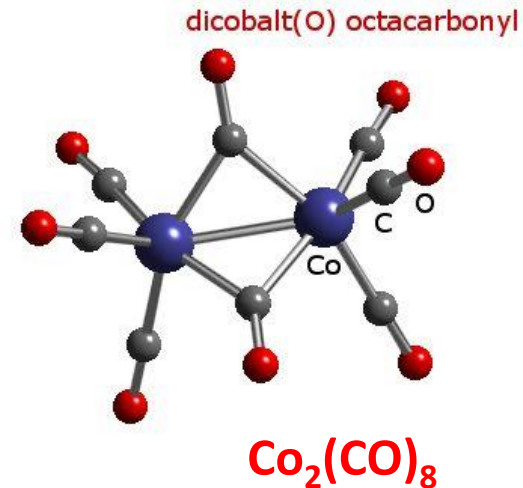


Example of organometallic compounds

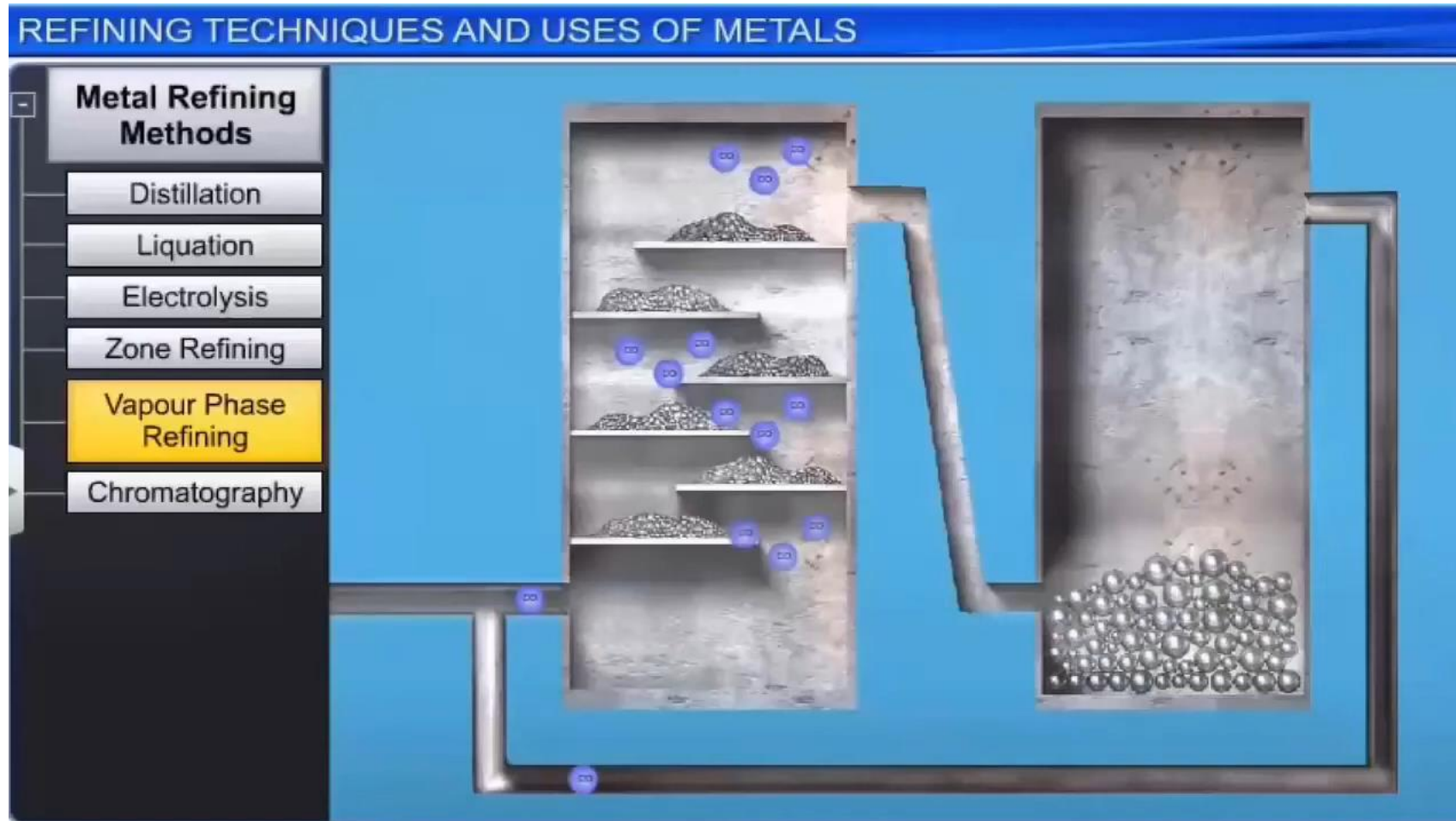
An area which bridges organic and inorganic chemistry

A branch of coordination chemistry where the complex has one or more metal-carbon bonds

M-C bond can be a σ type or π type bond



Mond Process for Purification of Nickel : $\text{Ni}(\text{CO})_4$

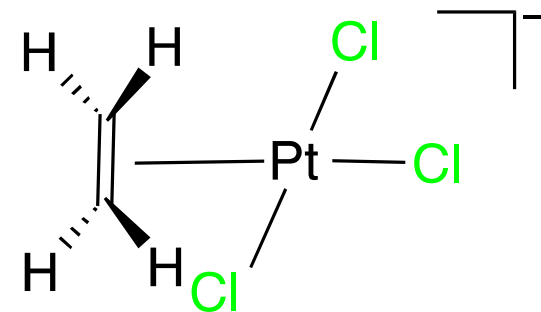
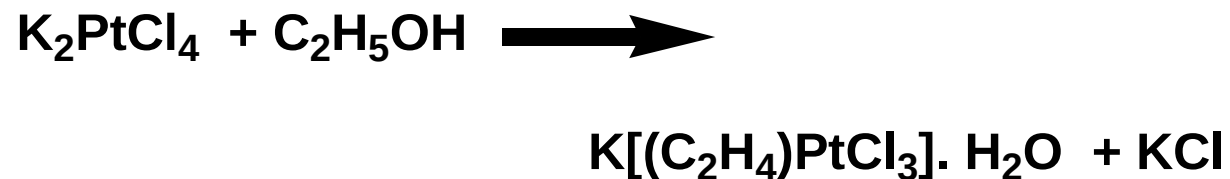


Zeise's Salt- The first transition metal organometallic compound



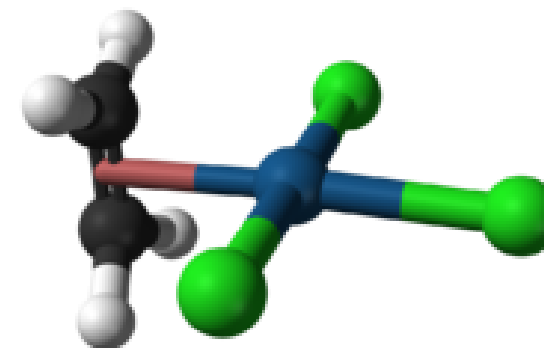
W C Zeise, Danish
pharmacist, **1789-
1847**

*Also father of the
chemistry of
mercaptans R-SH*



Discovery 1827

**Structure ~ 150
years later (1975)**

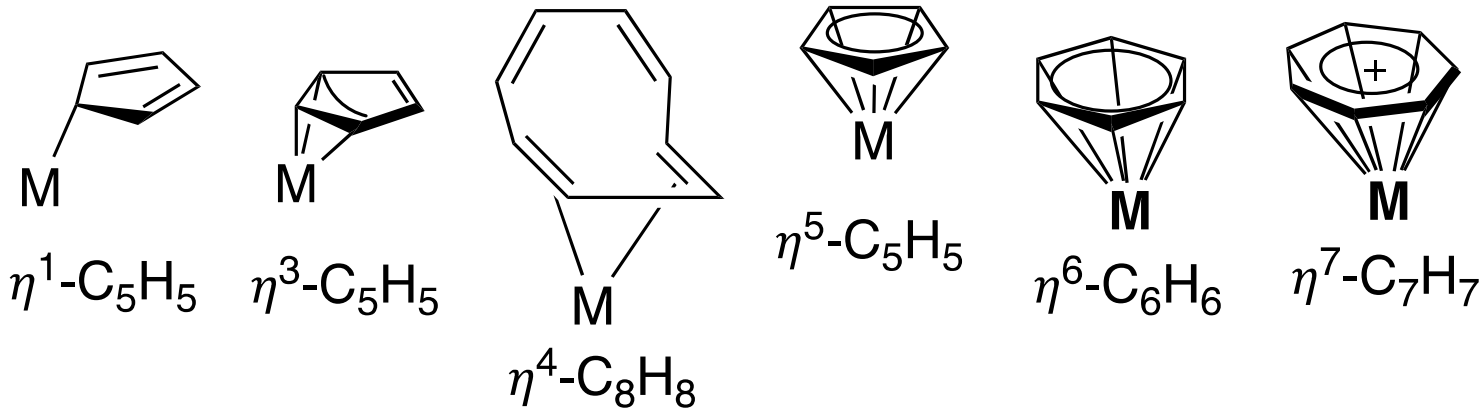


The breakthrough, the isolation of a pure, crystalline compound came when Zeise refluxed K_2PtCl_4 in ethanol and evaporated the resulting solution. Beautiful lemon yellow crystals, often one half inch or more in length were isolated. **The choice of Platinum was fortunate!!**

Chemists in those days often reported how the compounds that they had prepared tasted. Zeise described the taste of this potassium salt as metallic, astringent and long lasting.'

Terminology of Ligands

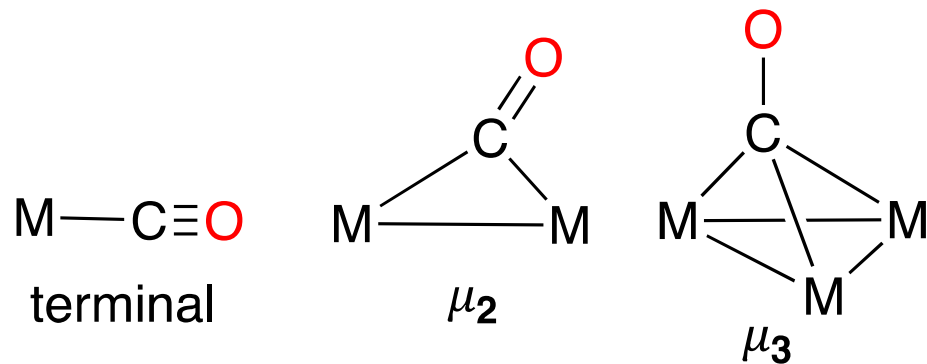
Hapticity of a ligand



The haptic symbol, η , with a numerical superscript, indicates the total number of donor atoms of a ligand attached to a metal centre.

For example, if all the five carbon atoms of a cyclopentadienyl moiety are equidistant from a metal atom, we term it as $\eta^5\text{-cyclopentadienyl}$

Bridging modes of CO: The symbol μ indicates bridging normally we have μ_2 and rarely μ_3 bridging

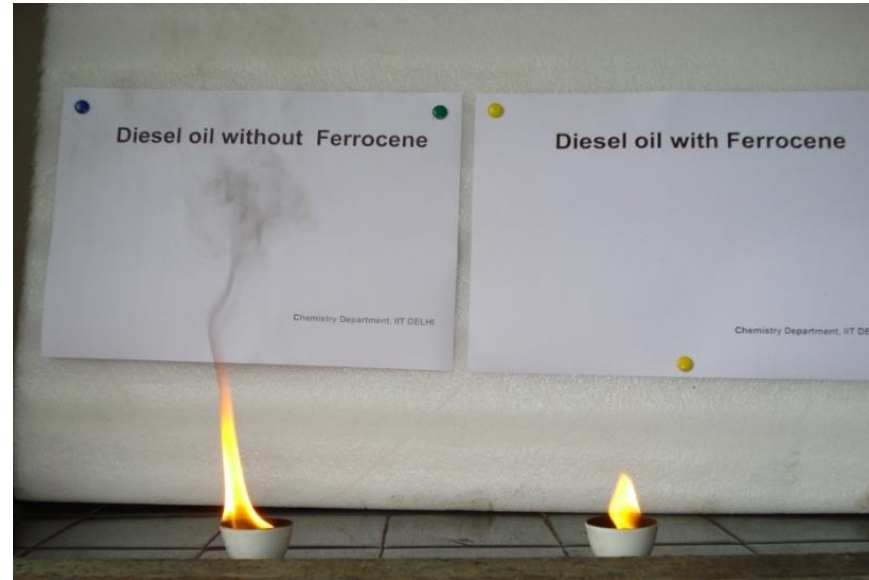


$\mu_2\text{-H}$, $\mu_2\text{-Cl}$, $\mu_3\text{-Cl}$, $\mu_2\text{-OR}$, $\mu_2\text{-PR}_2$, $\mu_2\text{-NR}_2$

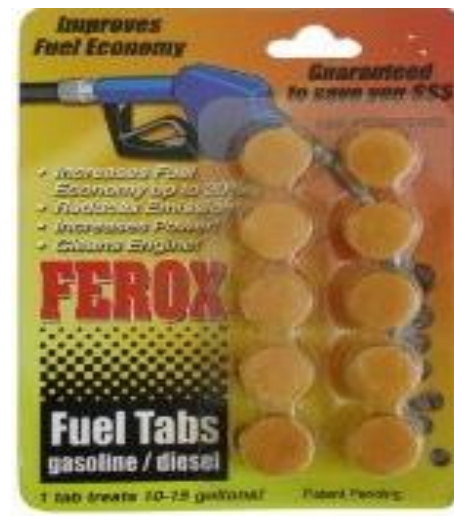
Ferrocene: Fuel additive, smoke suppressant and chiral catalyst precursor



Ferrocene powder



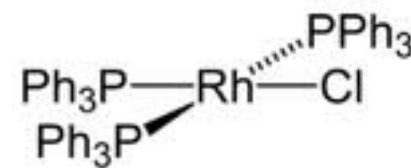
Ferrocene crystals



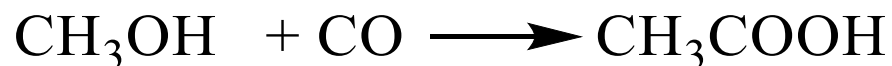
Ferox Gas & Diesel Fuel Additive is a catalyst that is an eco-friendly fuel additive and horsepower booster. It allegedly increases mileage from between 10 and 20% while also significantly reducing harmful emissions.

Organometallic catalysts in industrial synthesis : Three Nobel Prizes 2001, 2005 and 2010

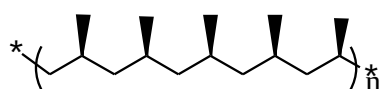
Hydrogenation



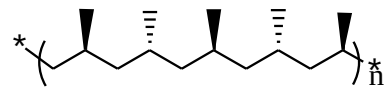
Methanol to acetic acid process



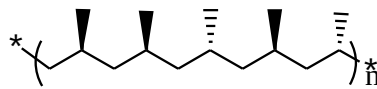
Olefin polymerization and oligomerization



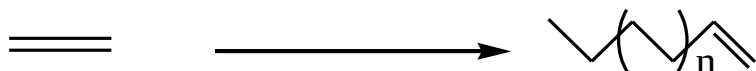
Isotactic polypropylene



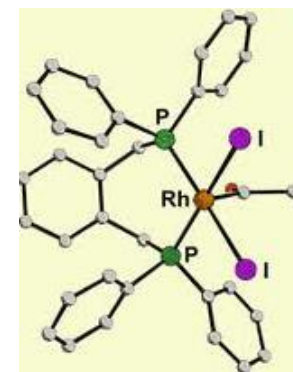
Syndiotactic polypropylene



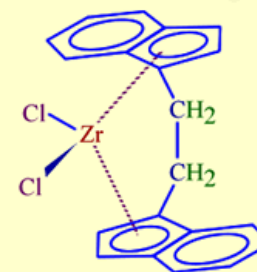
Atactic polypropylene



C ₄ -C ₈	40%
C ₁₀ - C ₁₈	40 %
C ₂₀ & >	20 %



A Metallocene Catalyst



On practical side, nearly 25 billion dollars was realized from industrial processes utilizing homogeneous catalysis based on organometallic chemistry in 1985. It was predicted that the role of organometallic chemistry in the production of pharmaceuticals, agrichemicals, flavours, fragrances and semiconductors will continue to expand in the next decades.

18 electron rule : How to count electrons



Nevil Vincent Sidgwick
(1873–1952)
British Chemist

The rule states that thermodynamically stable transition metal organometallic compounds are formed when the sum of the electrons on the metal plus the electrons donated from the ligands is equal to 36(Kr), 54 (Xe), 86 (Rn), the EAN rule was said to be obeyed.

An alternative and more general statement is that when a metal achieves an outer shell configuration of $ns^2 (n-1)d^{10} np^6$, there will be 18 electrons in the valence orbitals and a closed, stable configuration. This rule of thumb, which is referred to as the 18-electron rule

It has the advantage of being the same for all rows of the periodic chart, eliminating the need to remember a EAN for each noble gas. Furthermore, the number is an easy one to recall since it is merely the total capacity of nine orbitals, one set each of s, p, and d orbitals

Covalent Electron Counting Model

- This method uses the number of electrons that would be donated by ligands *if they were neutral*. For example chlorine ($\text{Cl}^\cdot$) considers as radical and it is 1e donor in this method.
- Formal oxidation state of metal considers as 0, in this method.



Fe atom	8 electrons
2(CO)	4 electrons
4 (Cl)	4 electron
2 -Charge	2 electron
Total	18 electrons

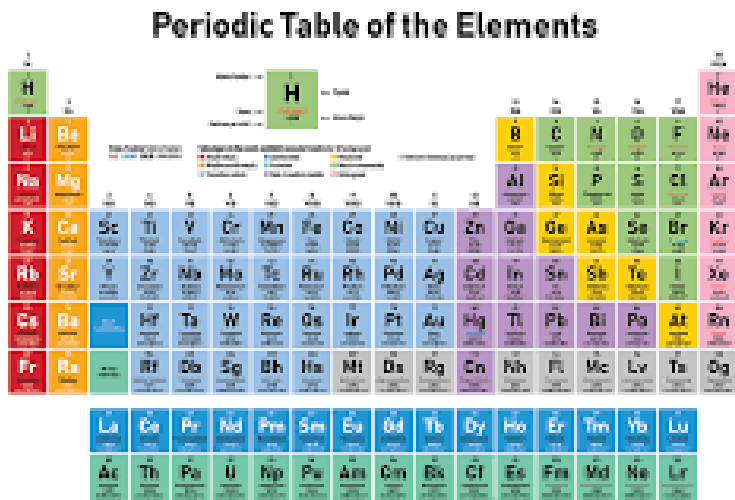
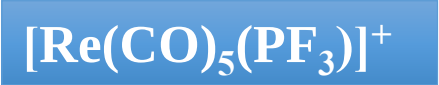
Ionic Electron Counting Model

- In the ionic model, each M–X is considered as arising from M^+ and X^- ions. For example, Cl^- considers as a 2e donor in this method . **Neutral ligands pose no problem because they are always 2e donors on either model.**
- To determine the total electron count, one must take into account the charge on each ligand and determine the formal oxidation state of the metal.



Fe(II) ion	6 electrons
2(CO)	4 electrons
4 (Cl^-)	8 electron
Total	18 electrons

Covalent Electron Counting Model



Re atom	7 electrons
5(CO)	10 electrons
1(PF ₃)	2 electron
1 +Charge	−1 electron
Total	18 electrons

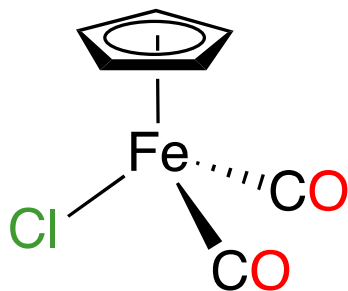
Ionic Electron Counting Model



Re(I) ion	6 electrons
5(CO)	10 electrons
1(PF ₃)	2 electron
Total	18 electrons

Covalent Model

Fe atom	8 electrons
$\eta^5\text{-C}_5\text{H}_5$	5 electrons
2(CO)	4 electrons
Cl	1 electron
Total	18 electrons

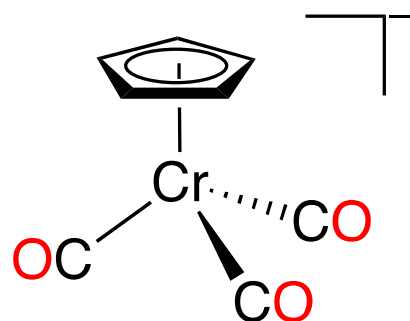


Ionic Model

Fe(II) ion	6 electrons
$\eta^5\text{-C}_5\text{H}_5^-$	6 electrons
2(CO)	4 electrons
Cl^-	2 electron
Total	18 electrons

Covalent Model

Cr atom	6 electrons
$\eta^5\text{-C}_5\text{H}_5$	5 electrons
3(CO)	6 electrons
1 $-$ Charge	1 electron
Total	18 electrons



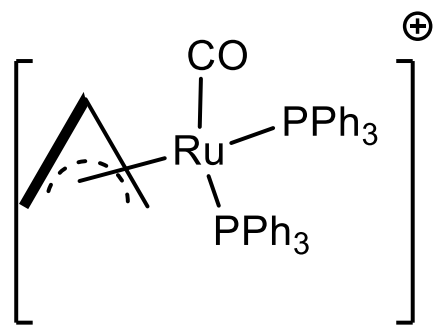
Ionic Model

Cr(0) ion	6 electrons
$\eta^5\text{-C}_5\text{H}_5^-$	6 electrons
3(CO)	6 electrons
Total	18 electrons

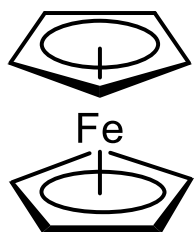
Neutral atom method: Metal is taken as in zero oxidation state for counting purpose

Oxidation state method: We first arrive at the oxidation state of the metal by considering the number of anionic ligands present and overall charge of the complex

Suggestion: Focus on one counting method till you are confident



	neutral atom method	oxidation state method
Ru	8	6 (Ru +2)
η^3 -allyl	3	4
2 PPh ₃	4	4
CO	2	2
charge	-1	not required
	16	16



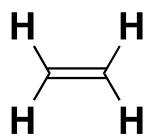
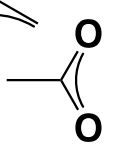
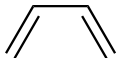
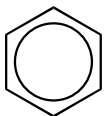
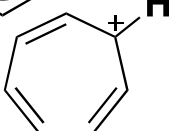
Fe	8	6 (Fe +2)
2 η^5 -Cp	10	12
	18	18

Neutral atom method: Metal is taken as in zero oxidation state for counting purpose

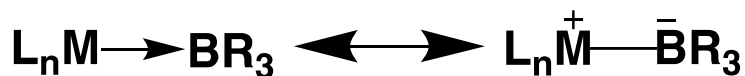
Oxidation state method: We first arrive at the oxidation state of the metal by considering the number of anionic ligands present and overall charge of the complex

Suggestion: Focus on one counting method till you are confident

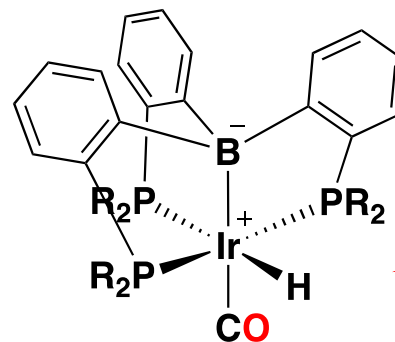
Electron Counting Schemes for Common Ligands

Ligands	Covalent Model	Ionic Model	Ligands	Covalent Model	Ionic Model
X = H, F, Cl, Br, I, OH, CN, CH ₃	1 (as X [•])	2 (as X ⁻)	Carbene, 	2	2
Lone-pair donors			Nitride, $\equiv\text{N}$	3	6 (N ³⁻)
CO, PR ₃ , NH ₃ , H ₂ O	2	2	=O, =S	2	4 (O ²⁻)
π -bond donors				-	-
	2	2			
σ -bond donors					
H-H	2	2			
η^3 -allyl, 	3	4			
η^2 -acetate, 	3	4			
η^4 -butadiene, 	4	4			
η^5 -cyclopentadienyl, 	5	6			
η^6 -benzene, 	6	6			
η^7 -tropylium ion, 	6	6			

Zero electron ligands



BR₃, having a 6e boron, completes its octet by accepting lone pairs, as in H₃N → BR₃ to become an 8e boron.



Donation of electron
From Ir filled *d* orbitals
to B *p* orbitals

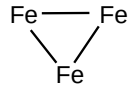
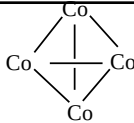
How to determine the total number of metal - metal bonds

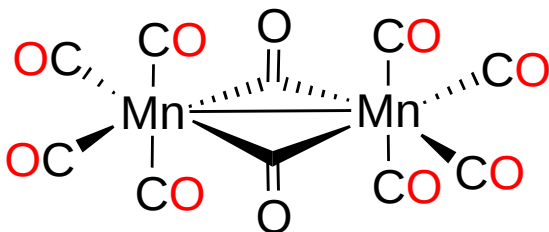
Determine the total valence electrons (TVE) in the entire molecule (that is, the number of valence electrons of the metal plus the number of electrons from each ligand and the charge); say, it is A .

Subtract this number from $n \times 18$ where n is the number of metals in the complex, that is, $(n \times 18) - A$; say, it is B .

(a) B divided by 2 gives the total number of M–M bonds in the complex.

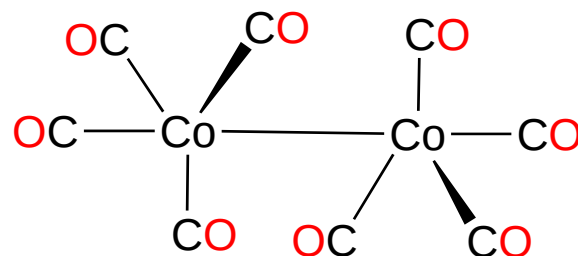
(b) A divided by n gives the number of electrons per metal. If the number of electrons is 18, it indicates that there is no M–M bond; if it is 17 electrons, it indicates that there is 1 M–M bond; if it is 16 electrons, it indicates that there are 2 M–M bonds and so on.

Molecule	TVE (A)	$(18 \times n) - A$ (B)	Total M–M bonds (B/2)	Bonds per metal	Basic geometry of metal atoms
$\text{Fe}_3(\text{CO})_{12}$	48	$54 - 48 = 6$	$6/2 = 3$	$48/3 = 16$; 2	
$\text{Co}_4(\text{CO})_{12}$	60	$72 - 60 = 12$	$12/2 = 6$	$60/4 = 15$; 3	
$[\eta^5\text{-CpMo}(\text{CO})_2]_2$	30	$36 - 30 = 6$	$6/2 = 3$	$30/2 = 15$; 3	$\text{Mo} \equiv \text{Mo}$
$(\eta^4\text{-C}_4\text{H}_4)_2\text{Fe}_2(\text{CO})_3$	30	$36 - 30 = 6$	$6/2 = 3$	$30/2 = 15$; 3	$\text{Fe} \equiv \text{Fe}$
$\text{Fe}_2(\text{CO})_9$	34	$36 - 34 = 2$	$2/2 = 1$	$34/2 = 17$; 1	$\text{Fe}-\text{Fe}$



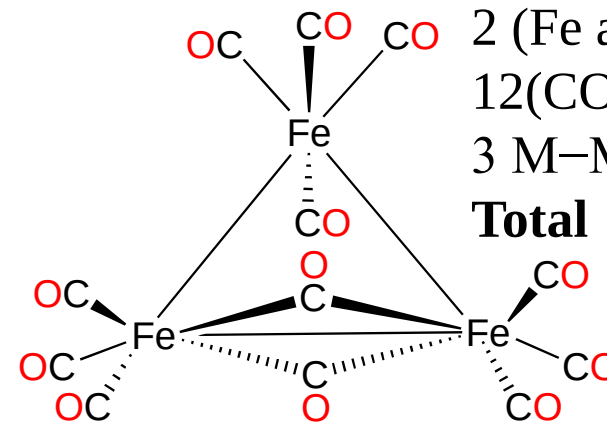
2 (Mn atom) 14 electrons
 10(CO) 20 electrons
 1 M–M 2 electron
Total **36 electrons**

18e⁻/Mn atom



2 (Co atom) 18 electrons
 10(CO) 16 electrons
 1 M–M 2 electron
Total **36 electrons**

18e⁻/Co atom



2 (Fe atom)
 12(CO)
 3 M–M
Total

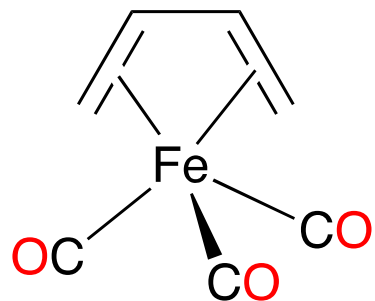
24 electrons
 24 electrons
 6 electron
36 electrons

18e⁻/Fe atom

Compound Name	Total valence electrons	e necessary to follow 18e rule	Number of M–M bond
$\text{Ru}_3(\text{CO})_{12}$	$(3 \times 8) + (12 \times 2) = 48$	$(3 \times 18) - 48 = 6$	3
$\text{Ir}_4(\text{CO})_{12}$	$(4 \times 9) + (12 \times 2) = 60$	$(4 \times 18) - 60 = 12$	6
$\text{Os}_4(\text{CO})_{16}$	$(4 \times 8) + (16 \times 2) = 64$	$(4 \times 18) - 64 = 8$	4
$\text{Rh}_6(\text{CO})_{16}$	$(6 \times 9) + (16 \times 2) = 86$	$(6 \times 18) - 86 = 22$	11

- **18-electron rule is very much useful to predict the number of M–M bond in a complex. However, it does not assist us to predict a terminal and bridging ligand.**

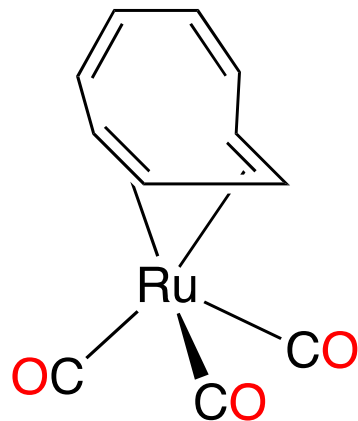
➤ Calculate the total number of valence electrons for each of the complexes.



[Fe(CO)₃(butadiene)]

Fe atom	8 electrons
η^4 -C ₄ H ₆	4 electrons
3(CO)	6 electrons

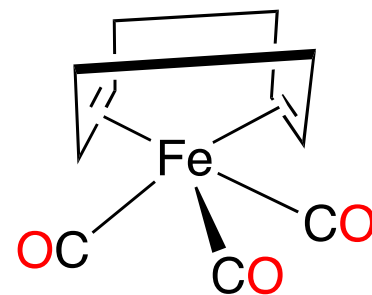
Total 18 electrons



[Ru(CO)₃(η^4 -COT)]

Ru atom	8 electrons
η^4 -COT	4 electrons
3(CO)	6 electrons

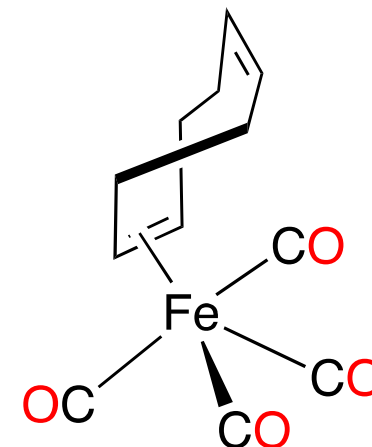
Total 18 electrons



[Fe(CO)₃(η^4 -COD)]

Fe atom	8 electrons
η^4 -COD	4 electrons
3(CO)	6 electrons

Total 18 electrons

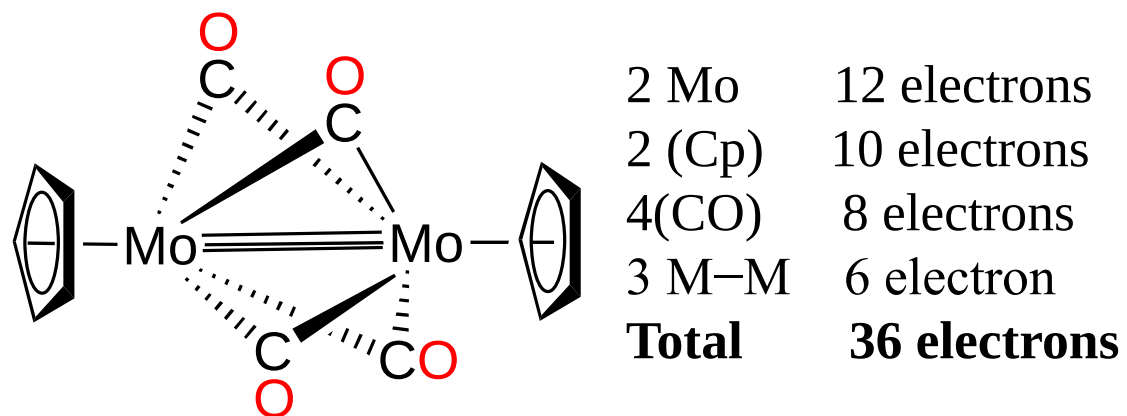
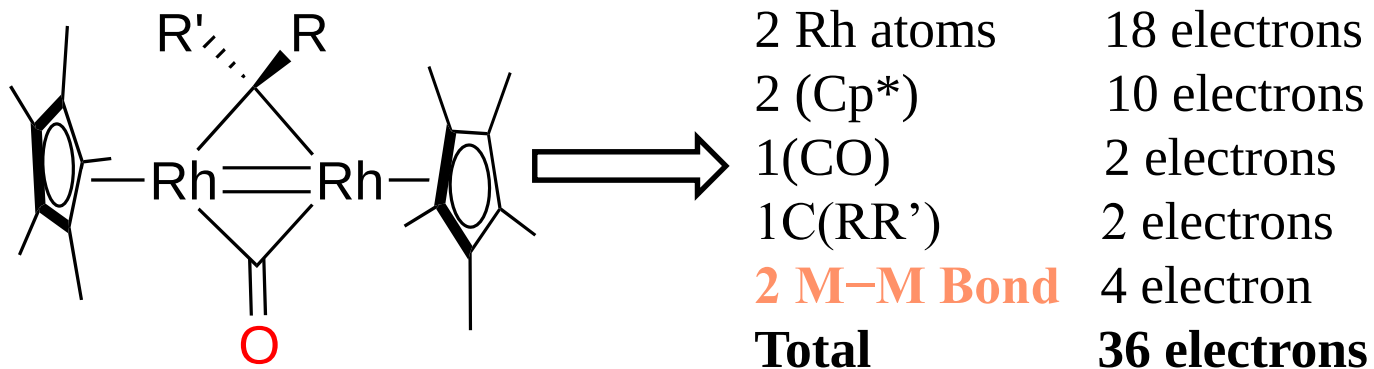


[Fe(CO)₃(η^2 -COD)]

Fe atom	8 electrons
η^2 -COD	2 electrons
3(CO)	6 electrons

Total 18 electrons

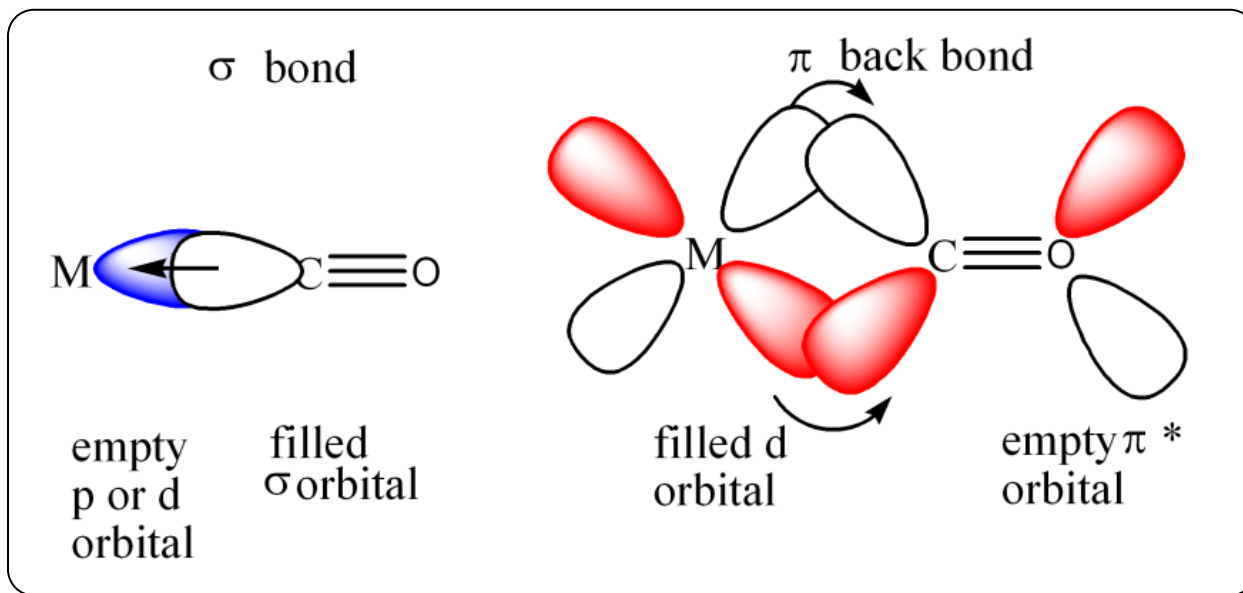
Find out the metal-metal bond in the following compounds?



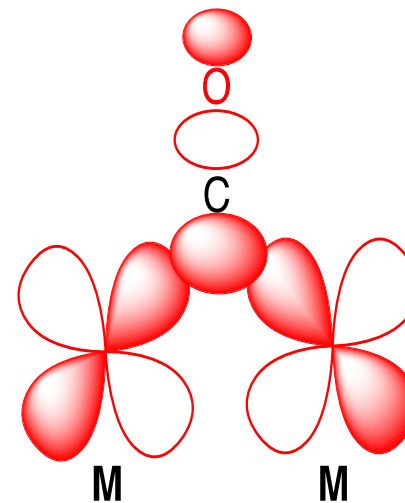
Find out the metal-metal bond in the following compounds?



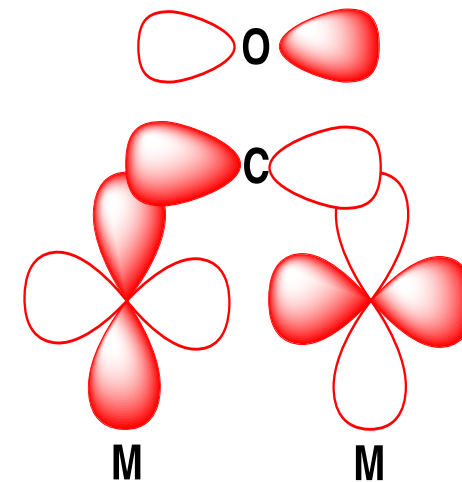
Binding mode of carbon monoxide



σ donation

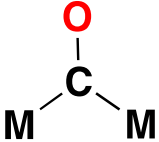
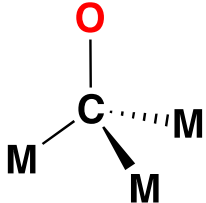


π acceptance



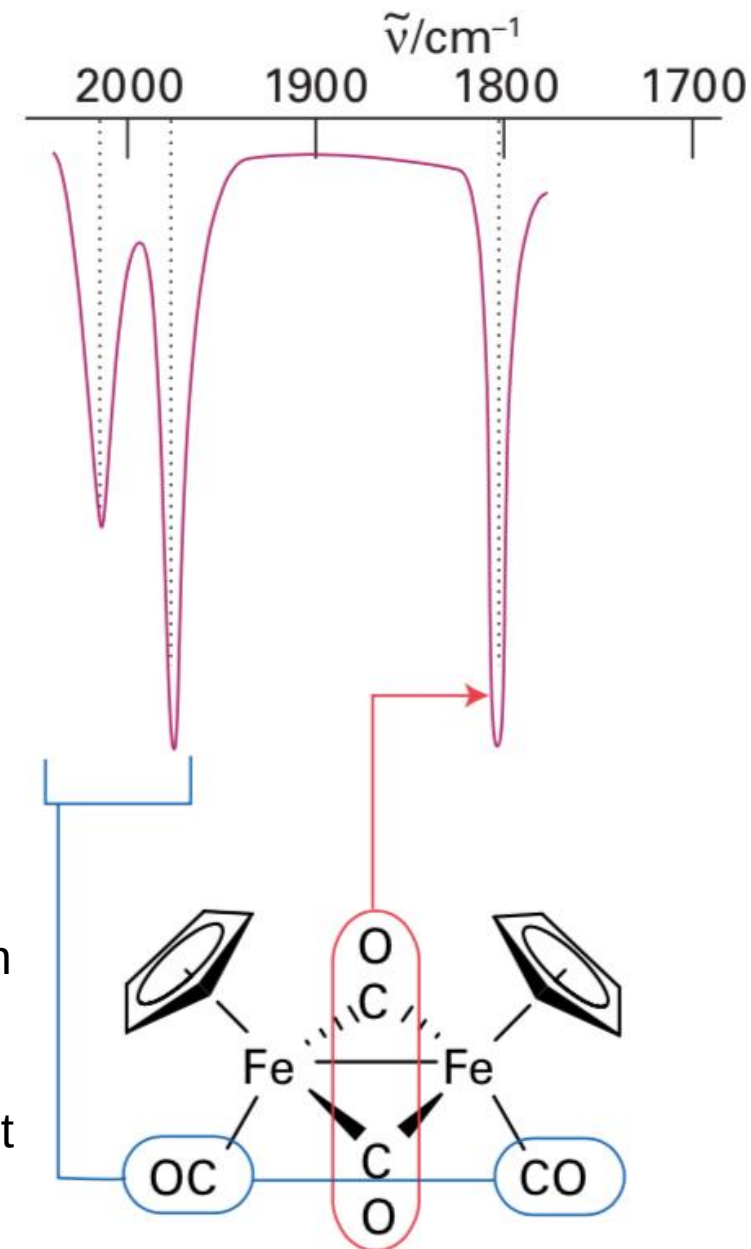
Counting the electrons helps to predict stability of metal carbonyls. But **it will not tell you whether a CO is bridging or terminal**

Bridging modes of CO

		Approximate range of $\nu_{\text{C-O}}$ in neutral complexes (cm^{-1})
Free CO		2143 cm^{-1}
Terminal C-O		1850-2120 cm^{-1}
Symmetrical $\mu_2\text{-CO}$		1700-1860 cm^{-1}
Symmetrical $\mu_3\text{-CO}$		1600-1700 cm^{-1}

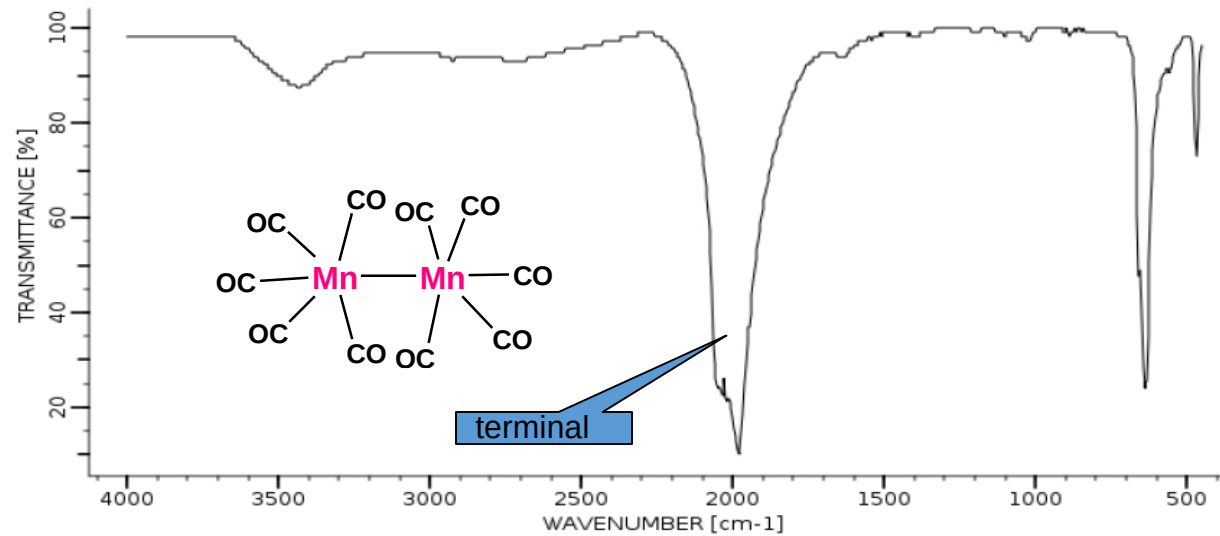
Characterization tools: Infrared spectroscopy and X-ray crystallography.

First, any change in the bonding between carbon and oxygen should be reflected in the C-O stretching vibration as observed by IR. The C-O stretch in organometallic complexes is often very intense (stretching the C-O bond results in a substantial change in dipole moment), and its energy often provides valuable information about the molecular structure. Free carbon monoxide has a C-O stretch at 2143 cm^{-1} . $\text{Cr}(\text{CO})_6$, on the other hand, has its C-O stretch at 2000 cm^{-1} . The lower energy for the stretching mode means that the C-O bond is weaker in $\text{Cr}(\text{CO})_6$.



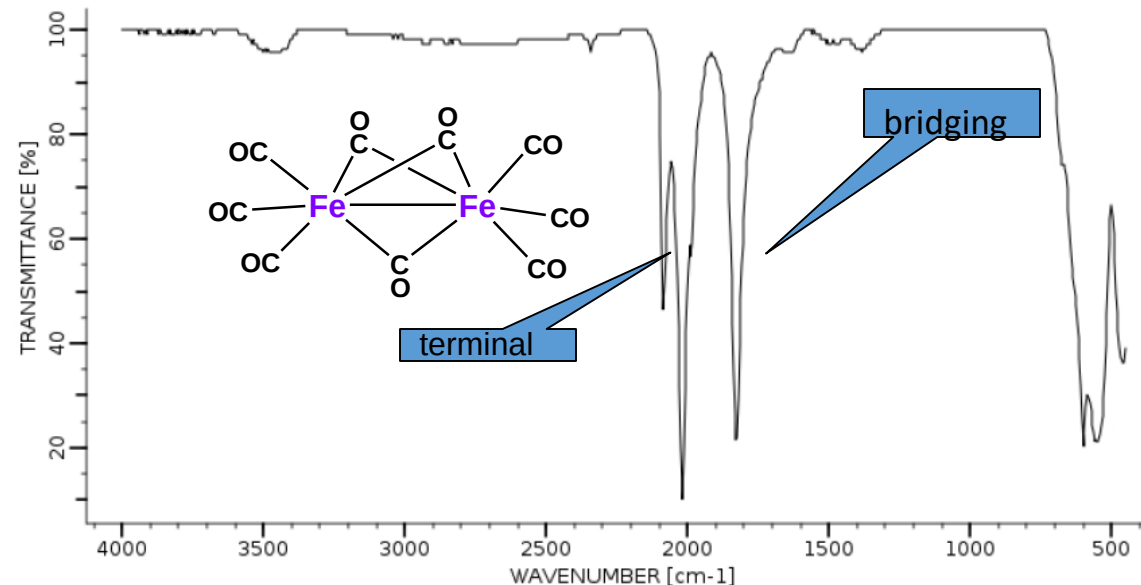
Infrared Spectroscopy- Spectra of Metal Carbonyls

1850-2120 cm^{-1}



The range in which the band appears decides bridging or terminal .

1700-1850 cm^{-1}

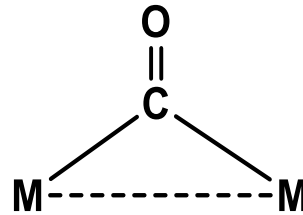


The number of bands is only related to the symmetry of the molecule

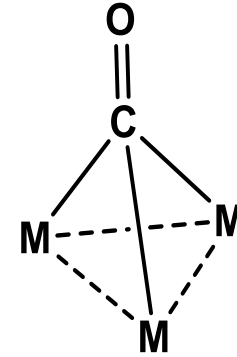
Terminal versus bridging carbonyls



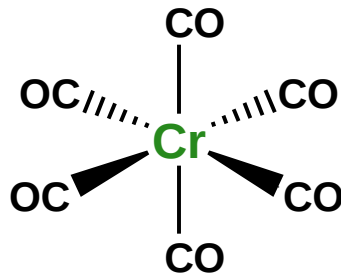
terminal
 ν_{CO} 2120-1850 cm^{-1}



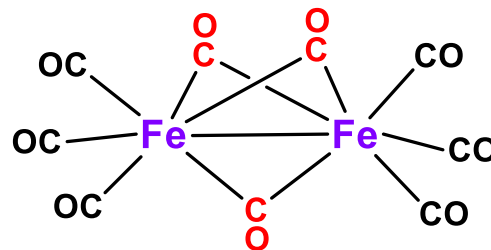
bridging μ_2
1850-1700 cm^{-1}



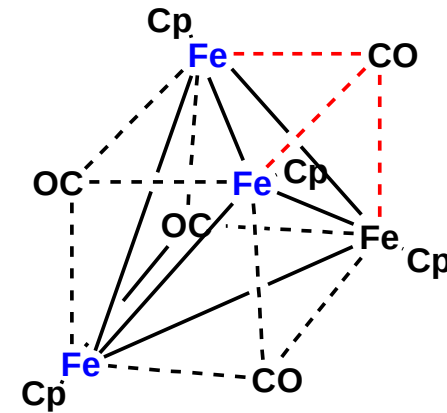
bridging μ_3
1730-1620 cm^{-1}



2000 cm^{-1}



2018, 1826 cm^{-1}



1620 cm^{-1}

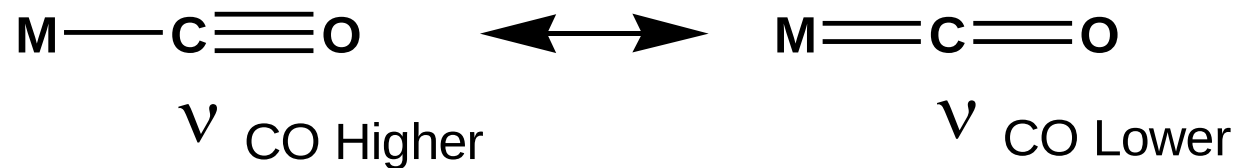
Factors which affect ν_{CO} stretching frequencies

Variation in ν_{CO} (cm^{-1}) of the first row transition metal carbonyls

	free CO 2143	
Ni(CO)_4 2057		
Co(CO)_4^- 1890		$\text{Co}_2(\text{CO})_8$ 2044(av, ter)
$[\text{Fe(CO)}_4]^{2-}$ 1815		Fe(CO)_5 2030
$[\text{Mn(CO)}_4]^{3-}$ 1600,1790	Mn(CO)_6^+ 2098	$\text{Mn}_2(\text{CO})_{10}$ 2013 (av)
$[\text{Cr(CO)}_4]^{4-}$ 1462,1657	Cr(CO)_6 2000	
	V(CO)_6^- 1860	V(CO)_6 1976
	Ti(CO)_6^{2-} 1747	

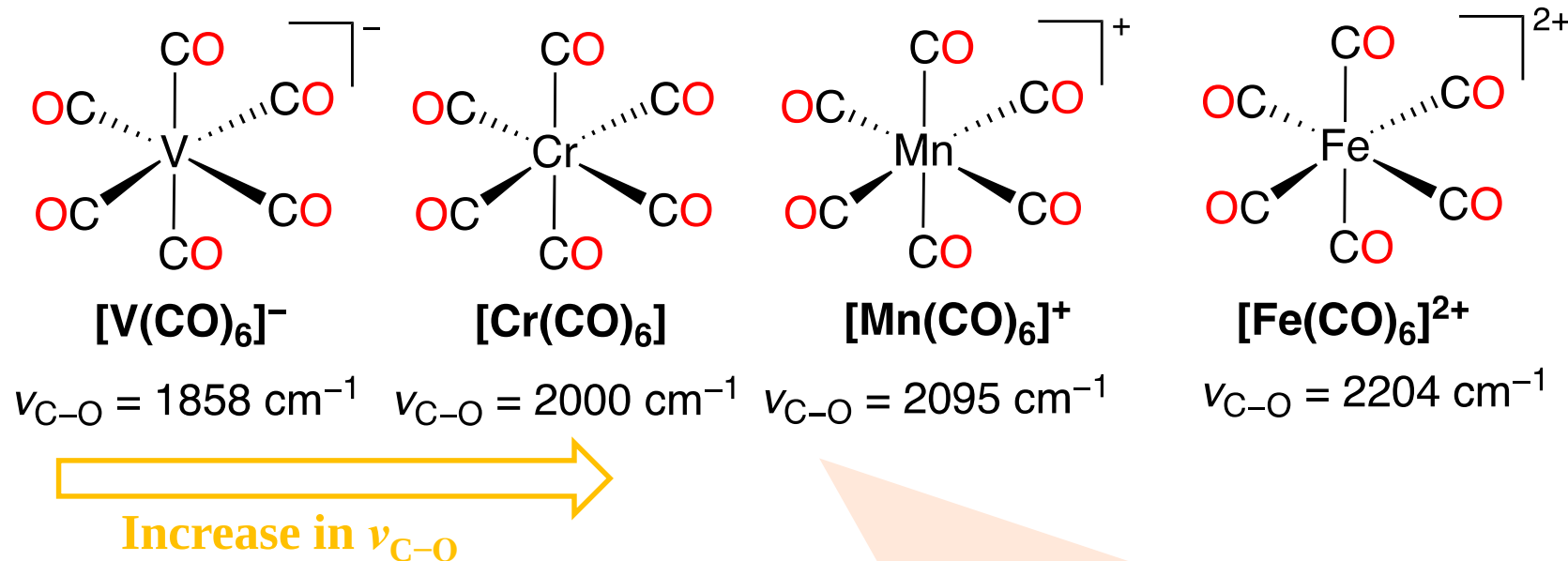
1. Charge on the metal
2. Effect of other ligands

As the electron density on a metal centre increases, more π -backbonding to the CO ligand(s) takes place. This weakens the C–O bond further as more electron density is pumped into the **empty π^* anti-bonding carbonyl orbital**. This increases the M–C bond order and reduces the C–O bond order. That is, the resonance structure $\text{M}=\text{C}=\text{O}$ becomes more dominant.

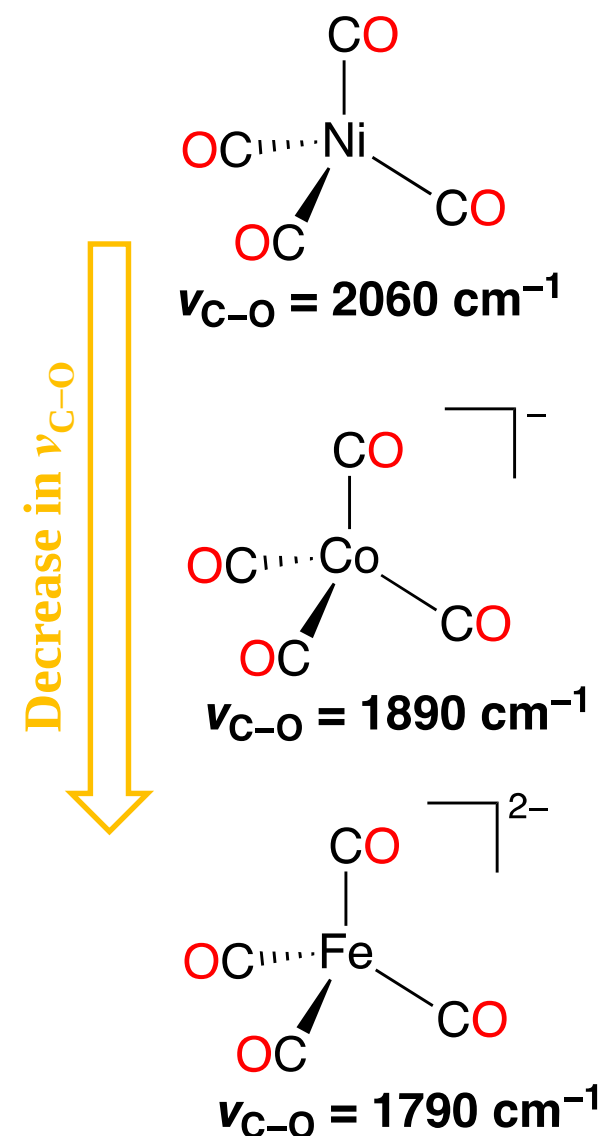


More back bonding

$\nu_{\text{C-O}}$ of some isoelectronic metal carbonyl complexes



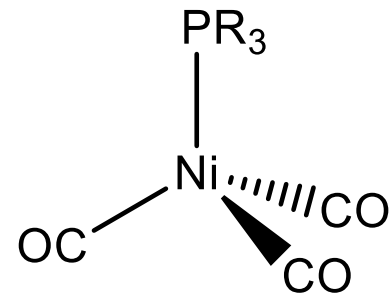
Of these four, $[\text{V}(\text{CO})_6]^-$ has the metal with the smallest nuclear charge. Consequently, vanadium has the weakest ability to attract electrons and the greatest tendency to “back” donate electron density to CO. The consequence is strong population of the π^* orbitals of CO and reduction of the strength of the C–O bond. In general, the more negative the charge on organometallic species, the greater the tendency of the metal to donate electrons to the π^* orbitals of CO and the lower the energy of the C–O stretching vibrations.



Other spectator ligands: Phosphines

PR_3	$\nu_{\text{CO}}, (\text{cm}^{-1})$	$\chi(\text{cm}^{-1})$ $\Delta \nu_{\text{CO}}$ wrt $\text{P}(\text{t-Bu})_3$	PR_3	$\nu_{\text{CO}}, (\text{cm}^{-1})$	$\chi(\text{cm}^{-1})$ $\Delta \nu_{\text{CO}}$ wrt $\text{P}(\text{t-Bu})_3$
$\text{P}(\text{t-Bu})_3$	2056.1	0.0	$\text{PPh}_2(\text{C}_6\text{F}_5)$	2074.8	18.7
PCy_3	2056.4	0.3	$\text{P}(\text{OEt})_3$	2076.3	20.2
$\text{P}(i\text{-Pr})_3$	2059.2	3.1	$\text{P}(p\text{-C}_6\text{H}_4\text{-CF}_3)_3$	2076.6	20.5
PEt_3	2061.7	5.6	$\text{P}(\text{OMe})_3$	2079.5	23.4
$\text{P}(\text{NMe}_2)_3$	2061.9	5.8	PH_3	2083.2	27.1
PMe_3	2064.1	8.0	$\text{P}(\text{OPh})_3$	2085.3	29.2
PBz_3	2066.4	10.3	$\text{P}(\text{C}_6\text{F}_5)_3$	2090.9	34.8
$\text{P}(o\text{-Tol})_3$	2066.6	10.5	PCl_3	2097.0	40.9
PPh_3	2068.9	12.8	PF_3	2110.8	54.7
PPh_2H	2073.3	17.2	$\text{P}(\text{CF}_3)_3$	2115.0	58.9

Lowest CO stretching frequency
Most donating phosphine
best σ donor



Highest CO stretching frequency
Least donating phosphine
best π acceptor

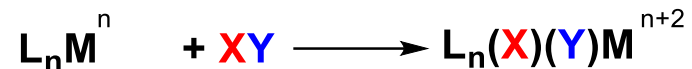
Organometallic Reaction Mechanism

Common paths in organometallic reactions

1. **Oxidative addition**
2. **Reductive elimination**
3. **Migratory Insertion**
4. **β -hydrogen elimination**
5. **σ -bond metathesis**

Oxidative addition

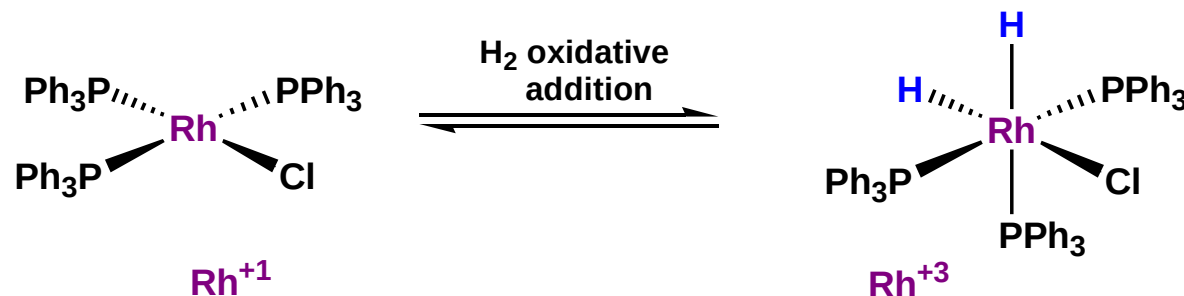
When addition of ligands is accompanied by oxidation of the metal, it is called an oxidative addition reaction



OX state of metal increases by 2 units

Coordination number increases by 2 units

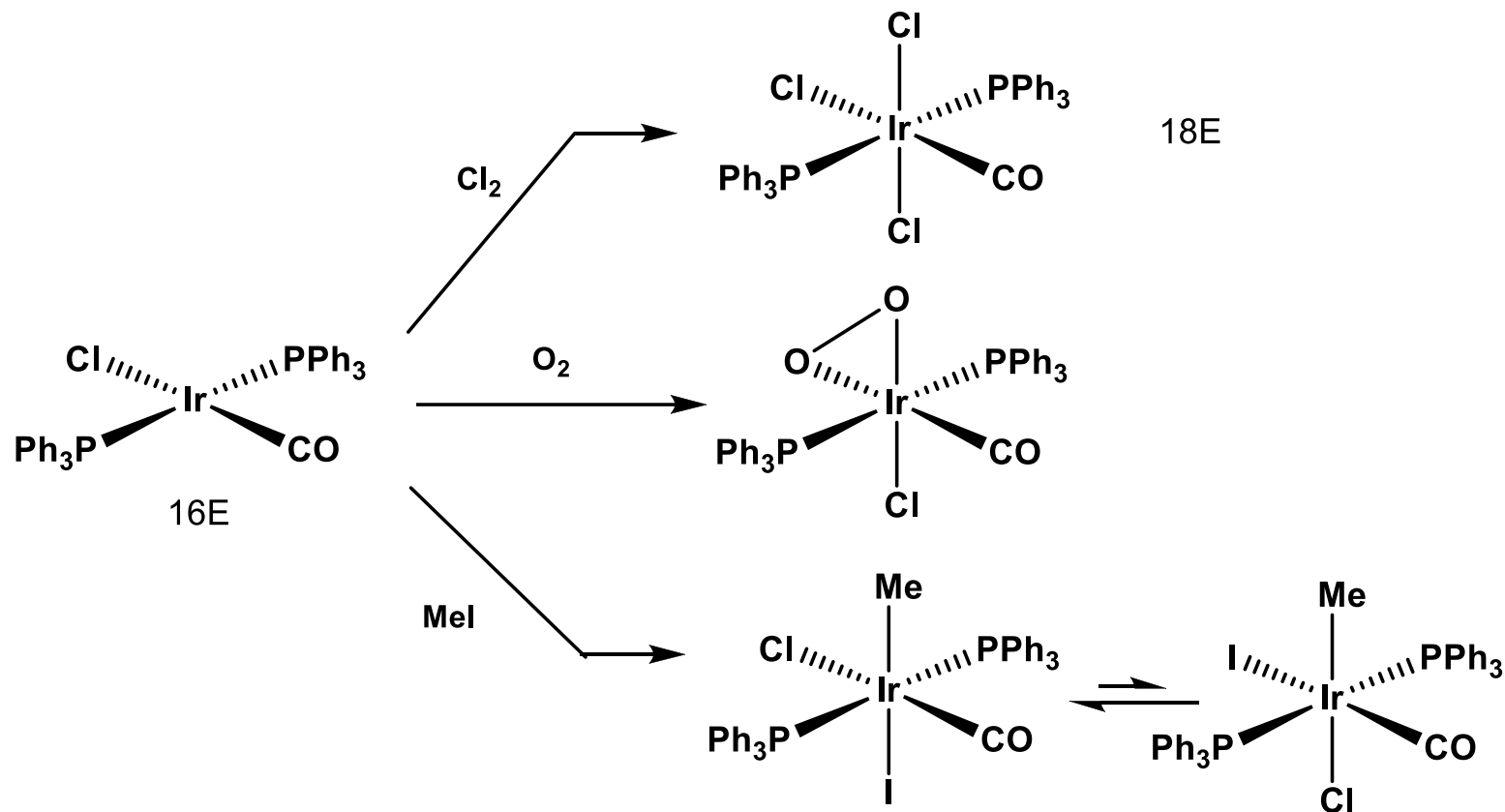
2 new anionic ligands are added to the metal



Requirements for oxidative addition

- Availability of nonbonded electron density on the metal
- Two vacant coordination sites on the reacting complex (L_nM), that is, the complex must be coordinatively unsaturated
- A metal with stable oxidation states separated by two units; the higher oxidation state must be energetically accessible and stable.

Examples of Oxidative addition : Cis or trans ?

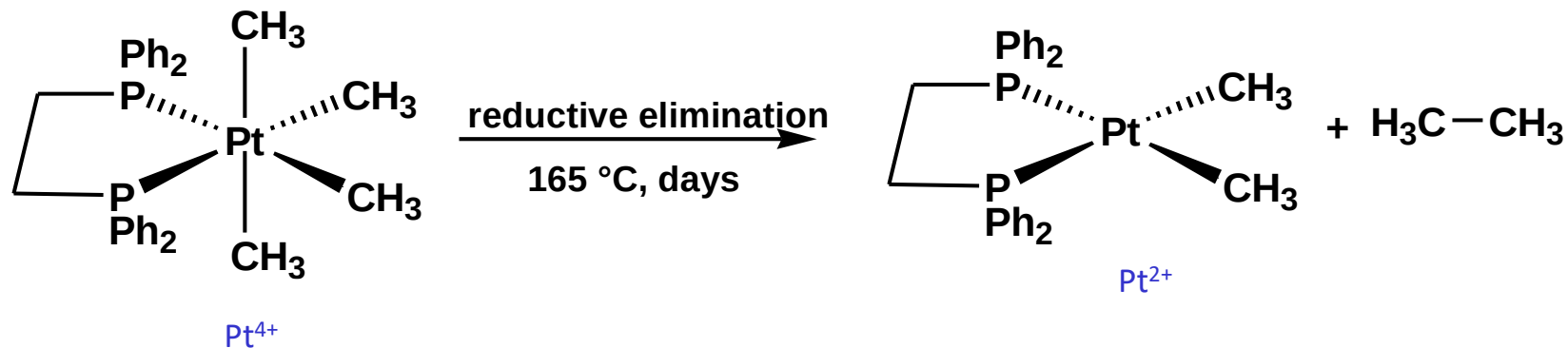


Homonuclear systems (H_2 , Cl_2 , O_2 , C_2H_2) Cis

Heteronuclear systems (MeI) Cis or trans

Reductive elimination

Almost the exact reverse of Oxidative Addition



Oxidation state of metal decreases by 2 units

Coordination number decreases by 2 units

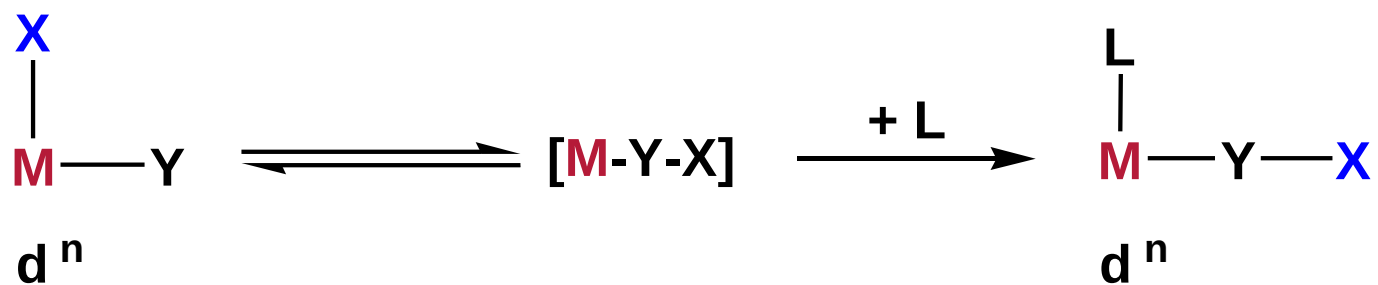
2 cis oriented anionic ligands form a stable σ bond and leave the metal

Factors which facilitate reductive elimination

- A high formal positive charge on the metal,
- The presence of bulky groups on the metal, and
- An electronically stable organic product.

Cis orientation of the groups taking part in reductive elimination is a MUST

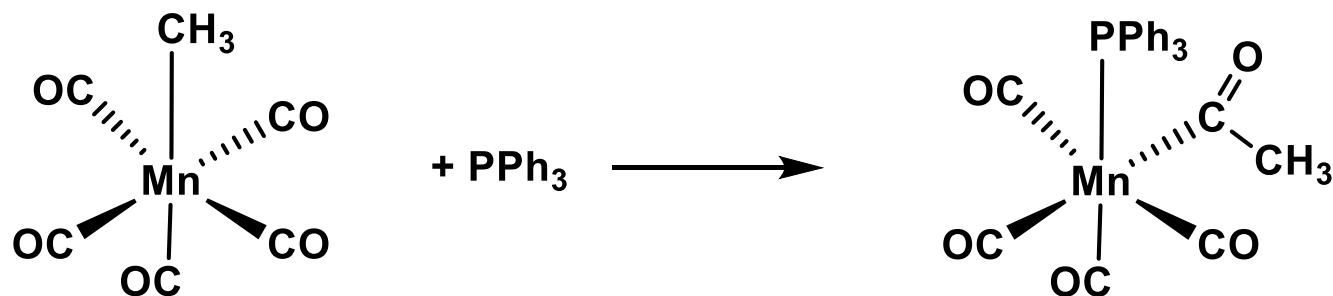
Migratory Insertion



No change in the formal oxidation state of the metal

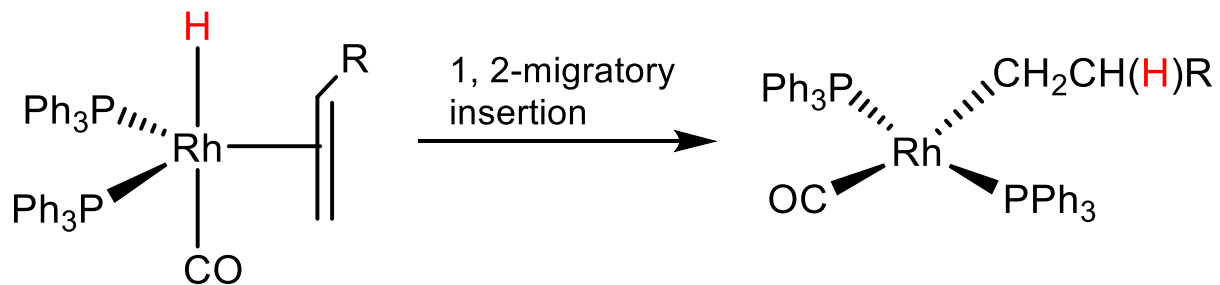
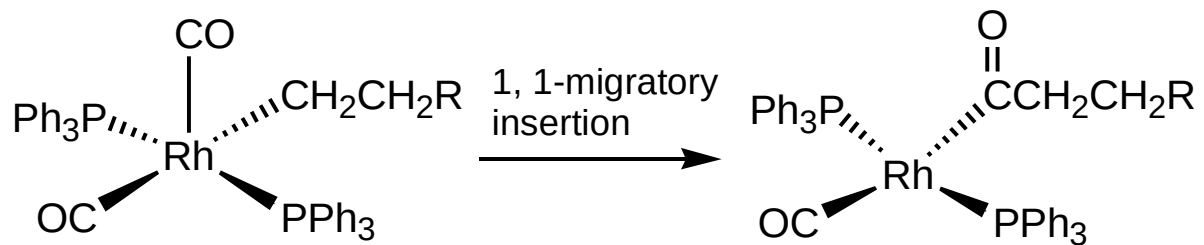
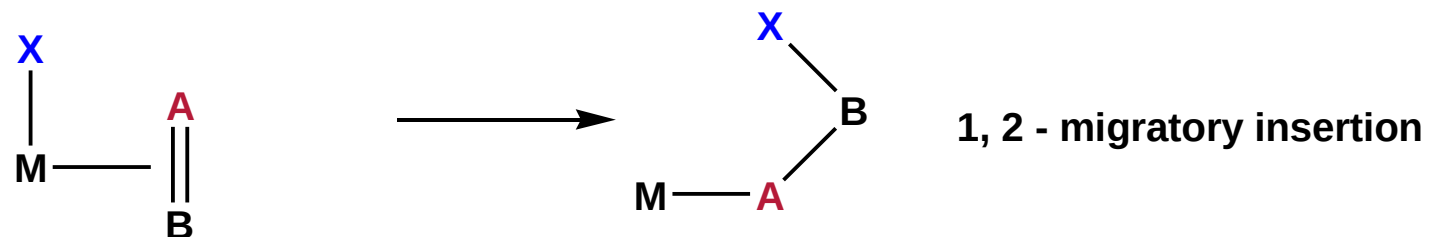
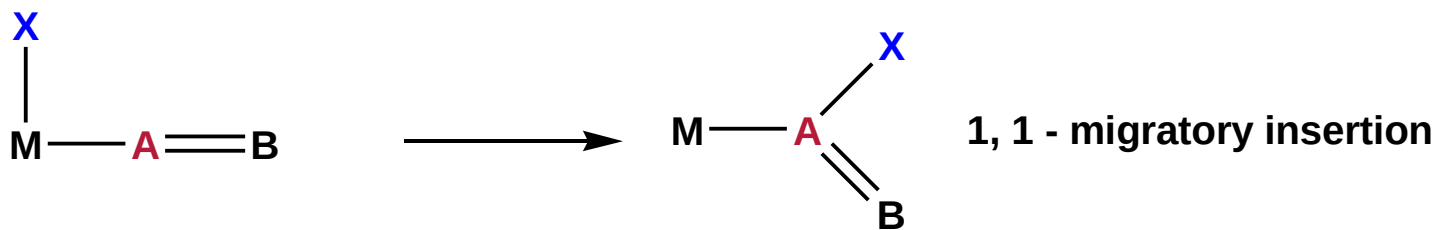
A vacant coordination site is generated during a migratory insertion (*which gets occupied by the incoming ligand*)

The groups undergoing migratory insertion **must be *cis*** to one another



These reactions are enthalpy driven and although the reaction is entropy prohibited the large enthalpy term dominates

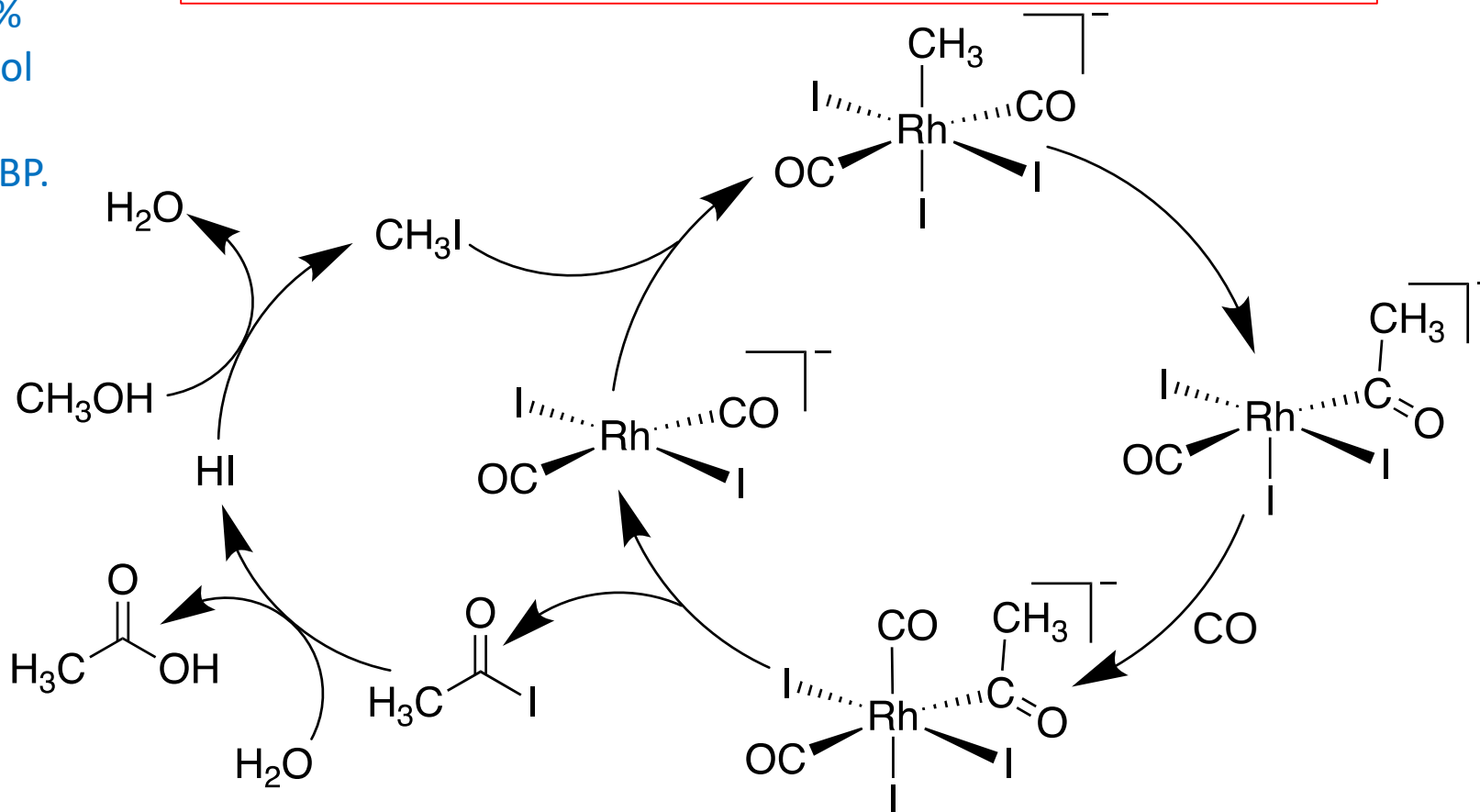
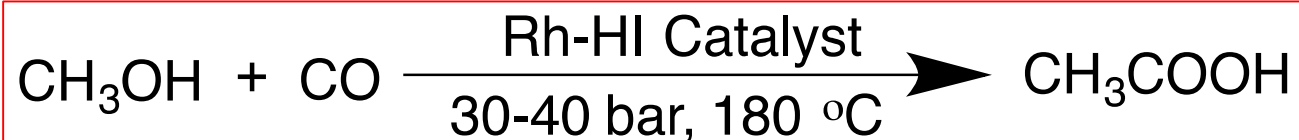
Types of Migratory Insertion



Homogeneous catalysis: industrial applications

Carbonylation of methanol: Monsanto Process

In 1966, Monsanto established that a rhodium based catalyst with iodide as a co-catalyst could give 99% conversion of methanol to acetic acid. This process was taken by BP.

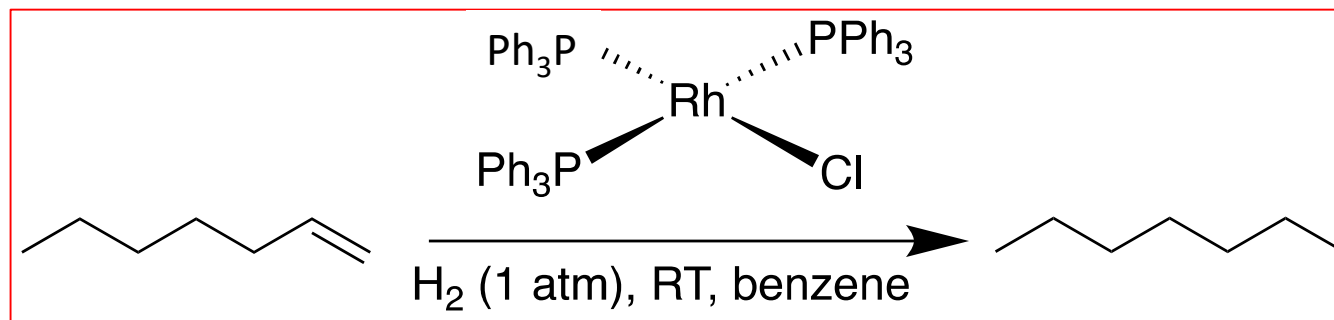


Most of Rh(III) salt and source of I^- , are suitable for the generation of active catalyst, $[\text{Rh}(\text{CO})_2\text{I}_2]^-$. OA of CH_3I to the later complex, is rate determining step in the reaction.

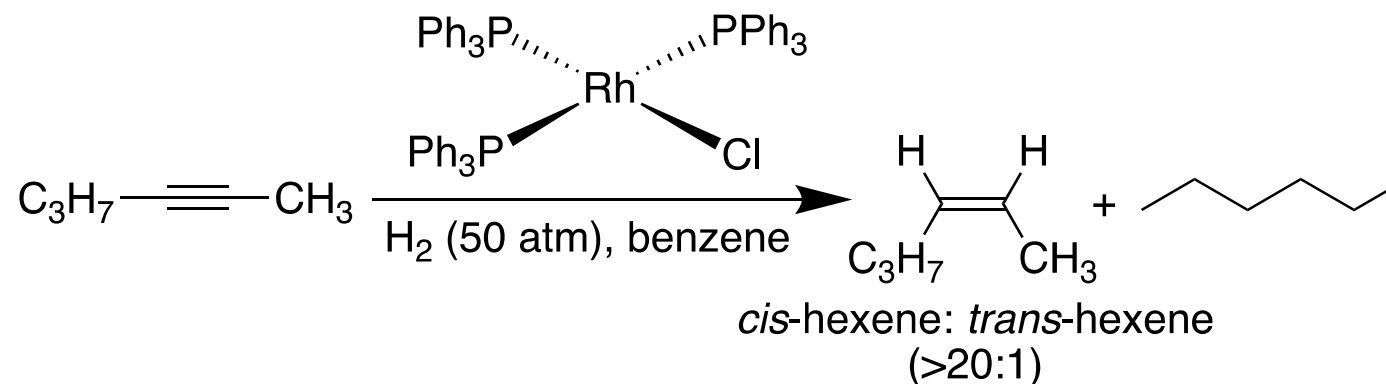
Wilkinson's Catalyst

Wilkinson's catalyst: The first example of an effective and rapid homogeneous catalyst for hydrogenation of alkenes, active at room temperature and atmospheric pressure.

Square planar 16 electron d^8 complex $(\text{Ph}_3\text{P})_3\text{RhCl}$

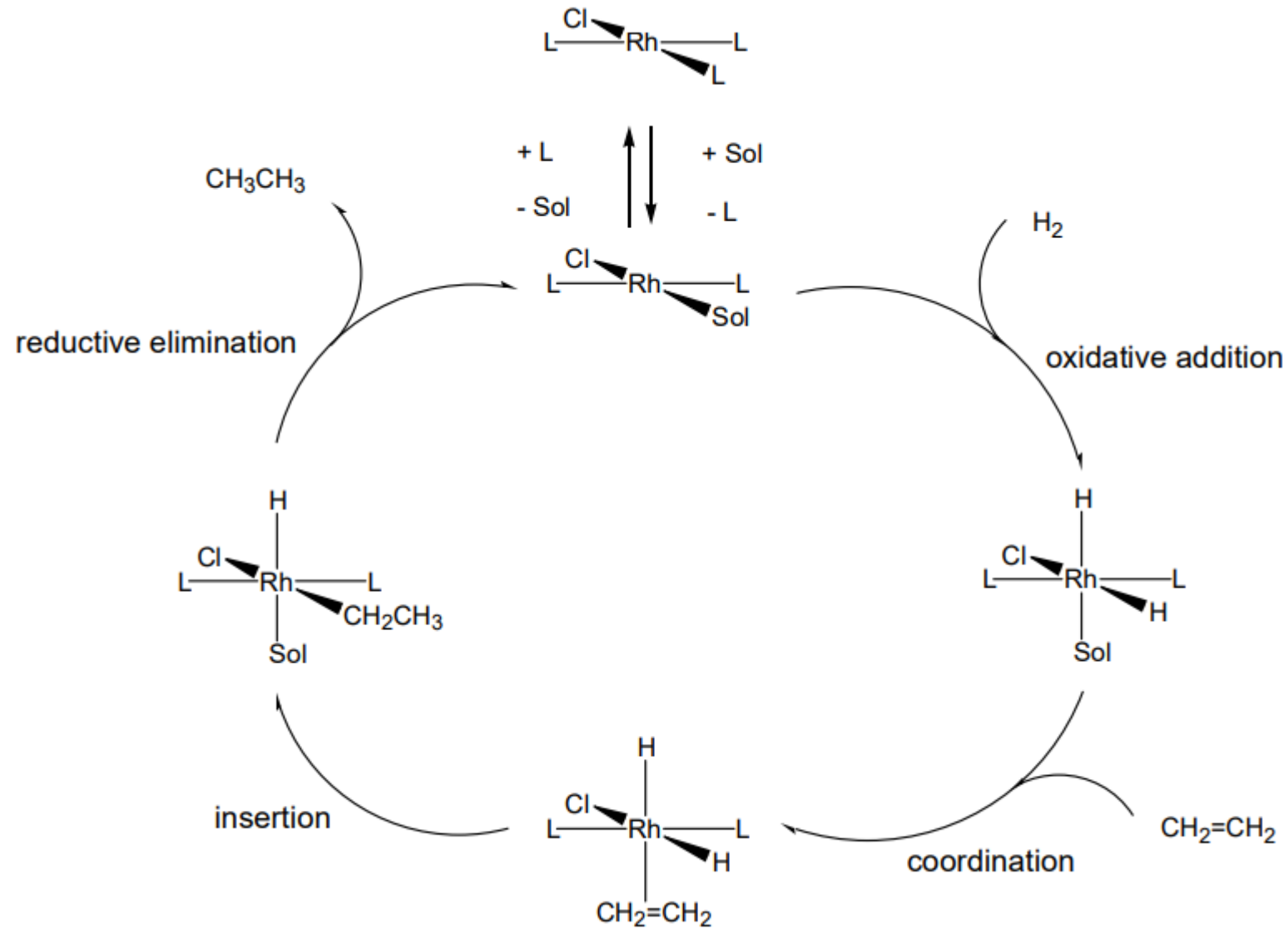


- These functional groups remain unaffected during hydrogenation reaction catalyzed by Wilkinson's catalyst.

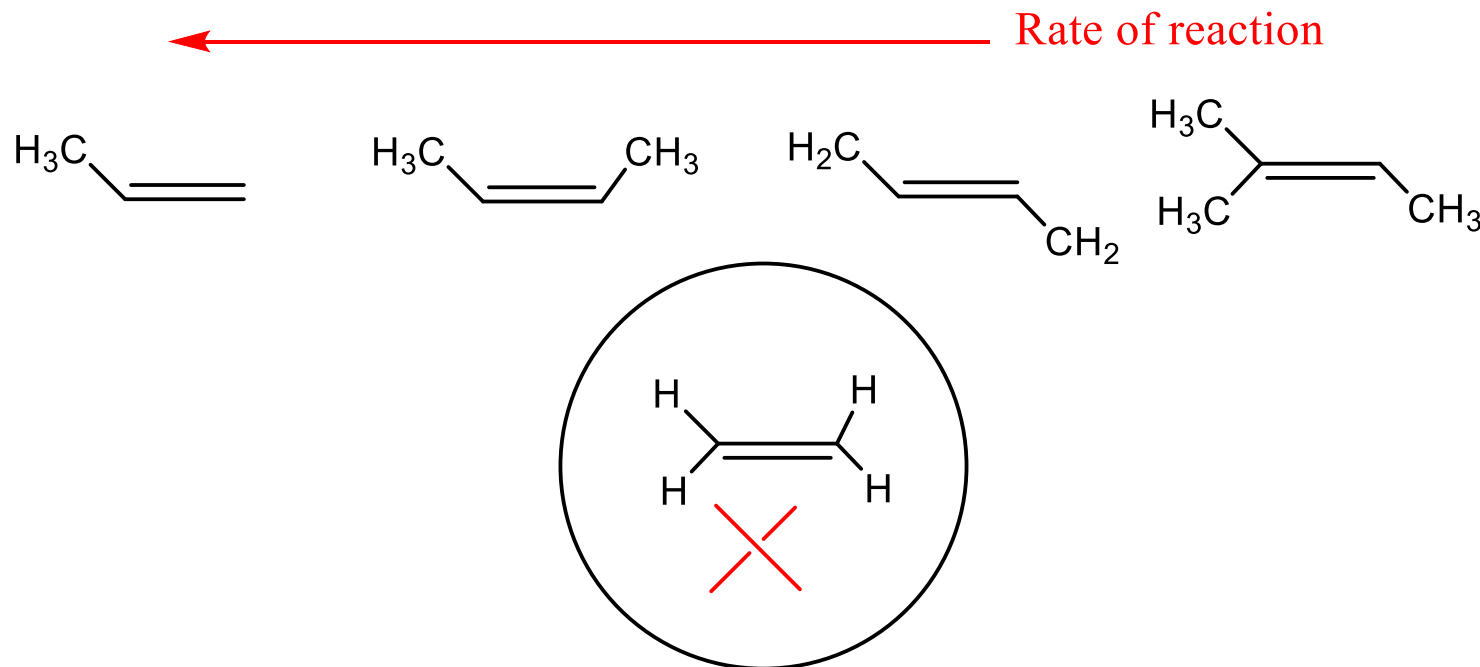


- Wilkinson hydrogenation reaction is stereospecific.

Catalytic cycle for hydrogenation



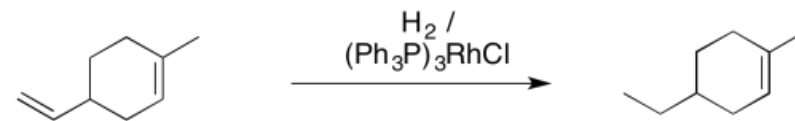
Find out the relative reactivity of the following alkenes for the homogenous catalytic hydrogenation with Wilkinson's catalyst

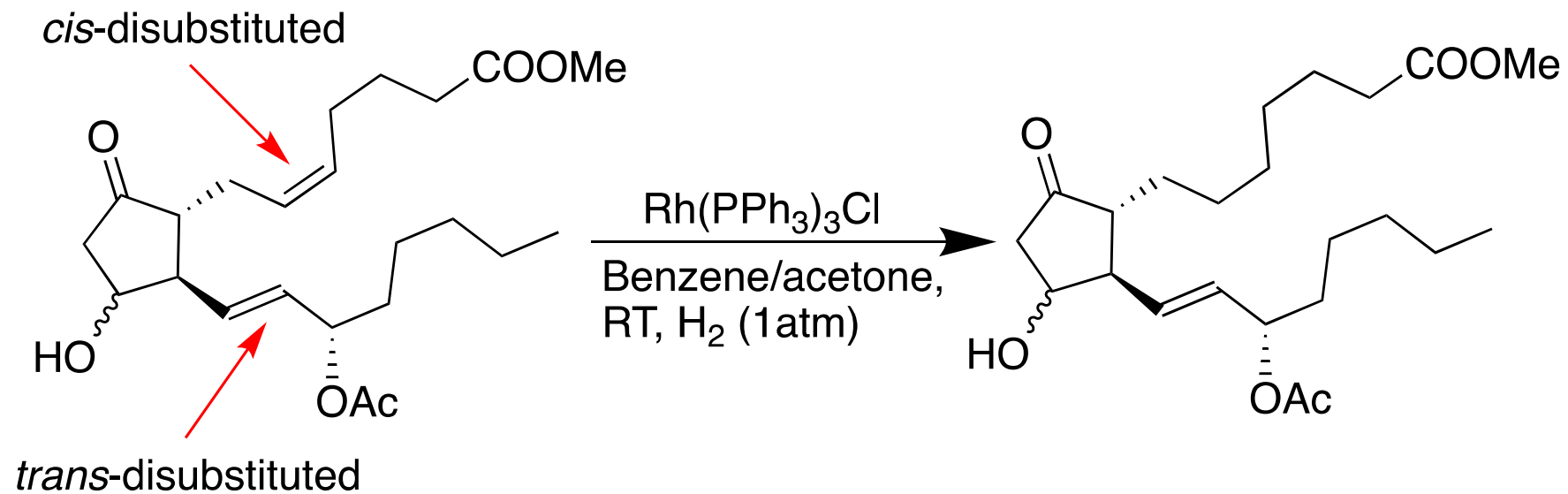
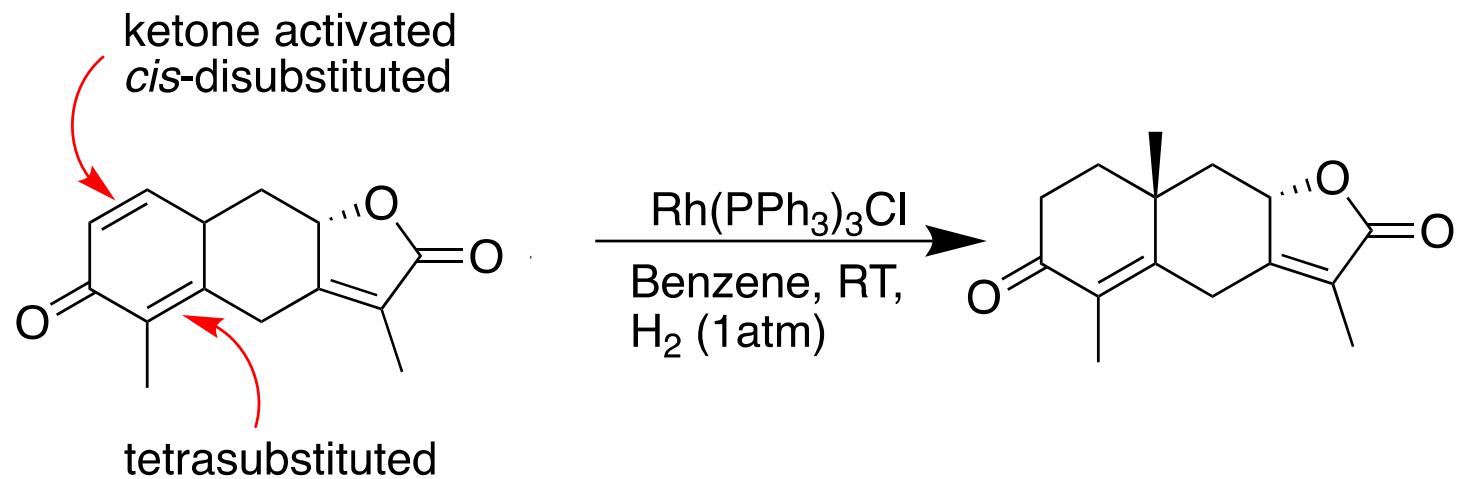


- Cis alkenes undergo hydrogenation more readily than trans alkenes

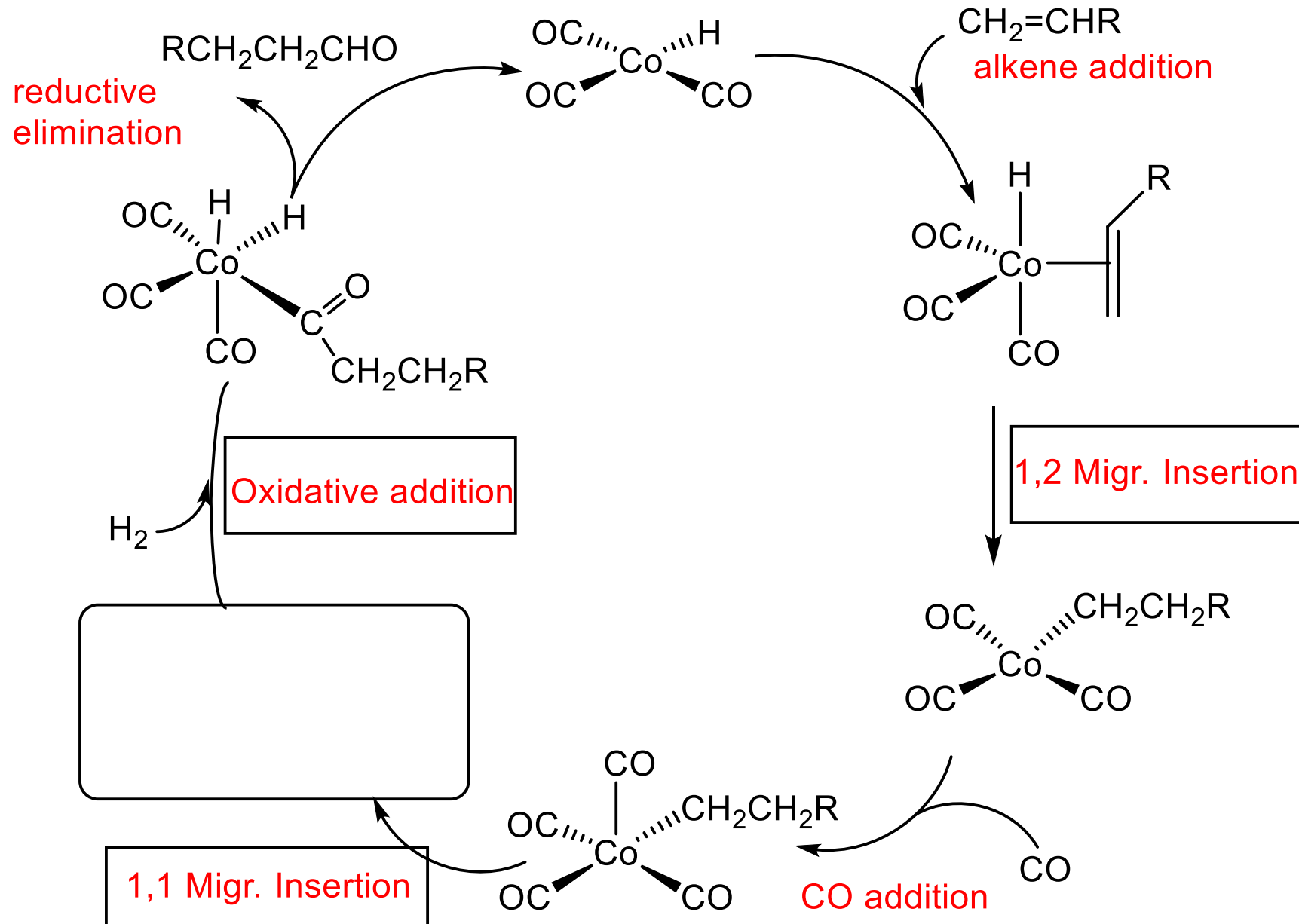
- Internal and branched alkenes undergo hydrogenation more slowly than terminal ones

- Strongly coordinating alkene such as ethylene binds preferentially Rh so that the resulting complex is stable and does not participate in the catalytic cycle.





Problem solving- fill in the blanks



Thank you